Representation Learning for Person Re-Identification: Baseline Reproduction and Variants on Market-1501 and DukeMTMC-reID

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ABSTRACT

Person re-identification (ReID) is a representation learning problem where a model must match the same identity across cameras under large appearance changes. In this report, we reproduce a strong baseline on Market-1501 and study variants on DukeMTMC-reID using the public PyTorch ReID baseline code. On Market-1501, the ResNet-50 baseline achieves Rank@1 = 0.8774 and mAP = 0.7219. We also observe a severe domain shift when directly testing the Market-trained model on DukeMTMC-reID (Rank@1 = 0.3299, mAP = 0.1700). Training on DukeMTMC-reID recovers performance (ResNet-50 Rank@1 = 0.7935, mAP = 0.6174), and using a stronger backbone (HRNet) further improves results (Rank@1 = 0.8389, mAP = 0.6946). Finally, we analyze two concrete failure cases and discuss improvement directions and AI safety considerations.

KEYWORDS

Person Re-identification, Representation Learning, Metric Learning, Retrieval, Deep Learning

1 INTRODUCTION

Person ReID aims to retrieve images of the same person identity from a gallery given a query image. Compared with closed-set classification, ReID is typically evaluated as a retrieval problem using embedding representations and distance-based ranking. This makes ReID a practical lens for studying representation learning, because good embeddings must be discriminative (separate different identities) and robust (invariant to camera, pose, occlusion, and illumination).

In this work, we reproduce a baseline model on Market-1501 [3] and run controlled variants on DukeMTMC-reID, covering baseline reproduction, cross-dataset domain shift, the impact of different backbones and loss functions, and failure-case analysis with safety reflection.

2 METHOD

2.1 Baseline architecture

We follow the codebase default design: a CNN backbone (ResNet-50 [1] by default) produces a feature map, followed by global pooling to obtain a fixed-length descriptor. The descriptor is projected to a 512-dimensional embedding and trained with an identity classification head (softmax cross-entropy).

2.2 Testing protocol

At test time, query and gallery embeddings are extracted and matched by distance. The evaluation reports Cumulative Matching Characteristics (CMC) and mean Average Precision (mAP). We also use a common test-time augmentation: horizontal flip, where features from the original and flipped image are averaged.

3 EXPERIMENTAL SETUP

3.1 Datasets

Market-1501 [3] is a widely-used ReID benchmark with 751 training identities and evaluation by query/gallery split. **DukeMTMC-reID** [2, 4] is another benchmark with 702 training identities and similar evaluation protocol.

3.2 Training configuration

Unless otherwise stated, we use the same training recipe across experiments for fair comparison: batch size 32, total epochs 60, initial learning rate 0.05, weight decay 5×10^{-4}, and dropout 0.5. The code is configured to train with –train_all, which merges the training and validation splits so that all labeled identities are used for supervision. This has two competing effects: the larger training set can improve feature coverage and generalization, but the absence of a held-out validation split removes the ability to monitor overfitting or perform early stopping based on retrieval performance. In practice, we observe that validation accuracy converges from 0.52 (epoch 0) to 0.98 (epoch 12) and remains stable thereafter, suggesting the model does not severely overfit within 60 epochs. However, without a principled validation signal, the choice of evaluation epoch (fixed at 60 throughout) is somewhat arbitrary. All performance comparisons therefore rely solely on the fixed query/gallery protocol rather than any held-out validation metric.

3.3 Commands and reproducibility

We use the following commands:

```
python train.py --gpu_ids 0 --name ft_ResNet50 --train_all
    --batchsize 32 --data_dir
    ./data/Market-1501-v15.09.15/pytorch
python test.py --gpu_ids 0 --name ft_ResNet50 --test_dir
    ./data/Market-1501-v15.09.15/pytorch --batchsize 32
    --which_epoch 060
python train.py --gpu_ids 0 --name duke_ft_ResNet50 --train_all
    --batchsize 32 --data_dir ./data/DukeMTMC-reID/pytorch
python train.py --gpu_ids 0 --name duke_ft_HR --train_all
    --batchsize 32 --data_dir ./data/DukeMTMC-reID/pytorch
    --use_hr
```

Model	Rank@1	Rank@5	Rank@10	mAP
ResNet-50 (softmax)	0.8774	0.9561	0.9724	0.7219

Table 1: Baseline results on Market-1501 using ft_ResNet50 **(epoch 60).**

4 RESULTS

4.1 Market-1501 baseline reproduction

Table 1 summarizes the reproduced baseline results on Market-1501.

4.2 Domain shift: Market-trained model on Duke

Directly applying the Market-trained model on Duke results in a large performance drop (Rank@1 = 0.3299, mAP = 0.1700), indicating strong domain shift. Compared with Market in-domain results (Rank@1 = 0.8774, mAP = 0.7219), the absolute drops are ΔRank@1 = 0.5475 and ΔmAP = 0.5519, both exceeding 50 percentage points. This is consistent with the intuition that camera networks, backgrounds, illumination, and clothing distribution differ across datasets, so representations trained for one domain may not transfer without adaptation. The magnitude of the drop also confirms that dataset-specific low-level statistics (e.g., color balance, background texture) are inadvertently encoded by the model alongside identity features.

Note that Rank@5 (= 0.4852) recovers noticeably relative to Rank@1 (= 0.3299). This suggests that correct gallery matches are not completely lost in the embedding space; rather, domain-specific hard negatives push the true match from rank 1 to ranks 2–5. The softmax loss optimizes inter-class separability but does not explicitly enforce intra-class compactness, so the learned embeddings carry an implicit dependence on the training domain's low-level statistics. When that distribution shifts, the relative ordering of same-identity pairs is disrupted more severely at the top of the ranking than further down.

4.3 Duke training and variants

Table 2 reports Duke results when training on Duke as well as variants. Among tested backbones, HRNet yields the best performance in our runs.

Backbone variants. DenseNet improves over the ResNet-50 baseline by +0.0216 Rank@1 and +0.0310 mAP, while HRNet achieves the largest gains (+0.0454 Rank@1, +0.0772 mAP), benefiting from its parallel multi-resolution representation that preserves spatial detail throughout training.

Loss function variants. Among the three alternative loss functions tested on the ResNet-50 backbone, Circle loss slightly underperforms the softmax baseline (−0.0107 Rank@1, −0.0046 mAP). Circle loss assigns an adaptive margin to each similarity score, which requires diverse and abundant per-identity samples to produce stable gradient signals; DukeMTMC-reID has fewer images per identity than Market-1501, so the circular decision boundary may not receive sufficient coverage to converge to a better optimum than softmax under the same 60-epoch budget. Instance loss recovers Rank@1 to 0.8007 (+0.0072 over baseline) while also improving mAP to 0.6232. By treating each image as its own class in a noise-contrastive objective, instance loss forces the model to distinguish fine-grained within-identity appearance variations rather than only coarse inter-identity boundaries, which benefits Rank@1 precision. Triplet loss achieves a similar mAP (0.6232) with Rank@1 at 0.7931, nearly matching the baseline. Both instance and triplet losses push same-identity pairs closer while pulling different-identity pairs apart, so their optimization objectives are structurally similar; the near-identical mAP values reflect that neither sampling strategy nor hard-mining was specially tuned in these runs, leaving the two methods with effectively equivalent discrimination power at this batch size. Overall, the gains from loss variants alone are modest compared with the gains from using a stronger backbone; combining HRNet with metric losses is a natural next step not covered here. Note that the current experiment matrix covers ResNet-50 × {softmax, circle, instance, triplet} and {DenseNet, HRNet} × softmax only; cross-combinations such as HRNet+triplet remain untested.

Setting	Rank@1	Rank@5	Rank@10	mAP
Market → Duke (zero-shot)	0.3299	0.4852	0.5485	0.1700
Duke baseline (ResNet-50)	0.7935	0.8896	0.9201	0.6174
Duke + DenseNet (–use_dense)	0.8151	0.9129	0.9367	0.6484
Duke + HRNet (–use_hr)	**0.8389**	**0.9241**	**0.9421**	**0.6946**
Duke + Circle loss (–circle)	0.7828	0.8842	0.9125	0.6128
Duke + Instance loss (–instance)	0.8007	0.8896	0.9183	0.6232
Duke + Triplet loss (–triplet)	0.7931	0.8878	0.9215	0.6232

Table 2: Results on DukeMTMC-reID (epoch 60). "Market → Duke" denotes testing Duke using the Market-trained ft_ResNet50 **without fine-tuning.**

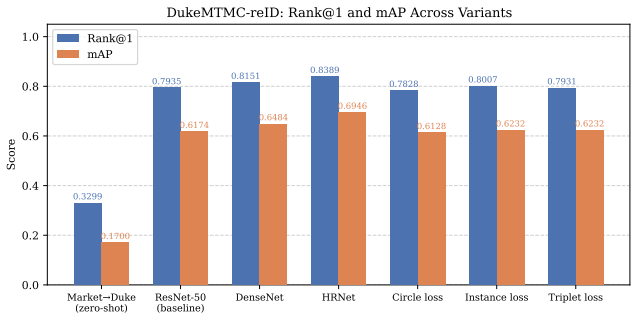

Figure 1: Rank@1 and mAP comparison across DukeMTMC-reID variants.

5 QUICK QUESTIONS

This section summarizes key practical questions from the tutorial and our experiments.

Why use AdaptiveAvgPool2d? Adaptive average pooling (e.g., output size 1 × 1) produces a fixed spatial output regardless of

Figure 2: Two DukeMTMC-reID success cases (Top-10 retrieval). Top: query index 0 (ID 5, cam 2). Bottom: query index 1 (ID 5, cam 5). All Top-10 results are correct matches.

Figure 3: Two DukeMTMC-reID failure cases (Top-10 retrieval visualization). Top: Case-1 (query index 1883, ID 4315). Bottom: Case-2 (query index 67, ID 0051).

the input resolution. This keeps the head shape-stable, while fixed-kernel average pooling would make the output depend on the input size.

Why horizontally flip images in test? Flip test-time augmentation averages features from original and flipped images. This reduces sensitivity to left-right pose bias and often improves retrieval robustness with minimal compute overhead.

Why L2-normalize features? L2-normalization puts embeddings on the unit hypersphere. Distances then emphasize angular differences and become less sensitive to scale, making similarity computation more stable for retrieval.

Why call `optimizer.zero_grad()`**?** Gradients accumulate by default in PyTorch. Without clearing old gradients each iteration, updates implicitly sum gradients across steps, which usually destabilizes training unless gradient accumulation is intended.

Why is the output dimension batch size $\times$ 751 **(or** 702**)?** The classification head outputs one logit per training identity, so the output width equals the number of training classes. Market-1501 has 751 training identities, yielding shape $N \times 751$; DukeMTMC-reID has 702 training identities, yielding $N \times 702$. This reflects that ReID training is framed as closed-set multi-class classification, even though test-time evaluation is open-set retrieval over unseen identities. At inference, the classification head is discarded and only the 512-dimensional embedding before the head is used for distance-based ranking.

6 FAILURE CASE ANALYSIS

For contrast, Figures 2 and 3 show two success cases and two failure cases from the Duke evaluation. In the success cases (Figure 2), all Top-10 retrieved images belong to the correct identity, indicating that the model produces compact, well-separated clusters for identities with consistent appearance across cameras. In the failure cases (Figure 3), both queries exhibit a "no positives in Top-10" pattern, where the first correct match appears extremely late in the ranking.

6.1 Case-1: missed match under cross-camera variation

Query image: 4315_c6_f0076814.jpg (ID 4315, query index 1883). The Top-1 retrieved image is an incorrect identity, 6367_c8_f0073570.jpg, and the first true match does not appear until rank 4274 (4315_c7_f0080881.jpg). This large rank gap indicates weak cross-camera invariance: viewpoint and illumination changes likely shift the query embedding away from its true cluster, while visually similar hard negatives occupy the top positions.

6.2 Case-2: false positives dominate Top-K

Query image: 0051_c1_f0060060.jpg (ID 0051, query index 67). The Top-1 result is also a false positive (6794_c8_f0178831.jpg), and the first correct gallery image appears only at rank 3277 (0051_c2_f0060553.jpg). The pattern suggests that the model over-relies on non-identity cues, such as background context or coarse clothing color, producing confident but semantically incorrect matches in early retrieval ranks.

7 IMPROVEMENT PROPOSALS

Based on the above results and observed failures, we propose:

- **Re-ranking:** apply k-reciprocal re-ranking [5] or GNN-based re-ranking to refine the initial ranking list. By encoding mutual nearest-neighbor relationships between query and gallery features, re-ranking can correct hard negatives near the decision boundary. In published benchmarks, mAP improvements of 3–8 percentage points are commonly reported on Duke-style datasets. The main cost is $O(N^2)$ pairwise distance computation and sensitivity to the hyperparameters k_1, k_2, and λ; these can be tuned on a held-out split. The codebase already provides `re_ranking.py` for direct use.
- **Stronger augmentation and sampling:** combine random erasing (`–erasing_p`, already available in the codebase), color jitter, and identity-balanced batch sampling to improve robustness to occlusion and illumination changes. Random erasing specifically addresses the occlusion patterns observed in Case-1 by forcing the model to use distributed, non-local cues rather than a single discriminative region. Identity-balanced sampling ensures that each mini-batch contains multiple images per identity, which is a prerequisite for effective metric loss training (triplet, circle) and also reduces the bias toward frequent identities.

- **Part-based pooling or attention modules:** insert a channel attention block (e.g., Squeeze-and-Excitation) or horizontal part pooling between the backbone feature map and the global descriptor. These modules direct the model to weight identity-relevant regions (head, torso, legs) over background, directly targeting the background-interference failure mode observed in Case-2. This direction is also consistent with the advantage of HRNet: by maintaining high-resolution spatial representations throughout training, HRNet implicitly retains part-level detail that global average pooling on lower-resolution maps discards. An explicit part or attention mechanism could provide similar benefits on a lighter backbone such as ResNet-50, reducing the compute overhead while recovering much of the spatial discriminability.

8 AI SAFETY REFLECTION

ReID is a dual-use technology. It can support positive applications such as search-and-rescue and authorized security auditing, but it can also enable non-consensual tracking and discriminatory surveillance.

To make the risks concrete, consider a shopping mall that deploys a ReID system to study pedestrian flow. If the embedding database is exfiltrated, an attacker can cross-reference it with other sources (loyalty-card records, parking data, or face recognition logs) to reconstruct fine-grained individual trajectories across days and locations without the subjects' knowledge. Even without a breach, the operator can silently repurpose the system from flow analytics to targeted individual monitoring, since the same embeddings support both use cases.

We suggest safeguards aligned with "lawful authorization + minimal necessity + auditability": minimize retention (store only necessary encrypted embeddings with bounded time-to-live), enforce access control and audit logs, apply rate limiting and similarity-score thresholding to external queries, and require independent governance reviews before high-risk deployments. Critically, operational purpose should be locked at system design time and enforced technically, not merely through policy.

9 CONCLUSION

We reproduced a strong ReID baseline on Market-1501 and evaluated transfer and variants on DukeMTMC-reID. The reproduced Market baseline achieves Rank@1 = 0.8774 and mAP = 0.7219. Zero-shot transfer from Market to Duke fails badly, highlighting domain shift. Training on Duke restores performance, and a stronger backbone (HRNet) improves retrieval quality further. Failure-case analysis suggests that cross-camera variation and hard negatives remain major challenges, motivating re-ranking and stronger augmentation as practical next steps.

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